Anil Kumar

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CAREER OBJECTIVE

AI / ML Engineer and M.Tech Data Science student at Amrita Vishwa Vidyapeetham, working across deep learning, computer vision, medical imaging, NLP, LLMs and agentic systems. Builds end-to-end PyTorch pipelines from dataset construction to training, honest evaluation and deployment. Seeking Machine Learning / AI / Computer Vision / GenAI roles.

TECHNICAL SKILLS

Languages & Libraries: Python, NumPy, Pandas, SciPy, scikit-learn, Matplotlib, Plotly, OpenCV

LLMs & Agents: Hugging Face Transformers, LangChain, LangGraph, Ollama, RAG, LLM-based labelling

Data & Streaming: DuckDB, PostgreSQL/PostGIS, PySpark, Kafka, Power BI, SQL

Deep Learning: PyTorch, TensorFlow, Keras, timm, CNN, LSTM/GRU, ViT, Swin, ConvNeXt, CLIP, AMP, CUDA

Machine Learning: XGBoost, LightGBM, CatBoost, Optuna, SHAP, ensembling, calibration, focal loss, TTA

Deployment & Cloud: AWS, Docker, Kubernetes, FastAPI, REST APIs, Git, Linux

EDUCATION

Amrita Vishwa Vidyapeetham, Bengaluru — *2025 – Present*
M.Tech in Data Science — *CGPA: 8.16*

Vel Tech R&D Institute of Science and Technology, Chennai — *2017 – 2021*
B.Tech in Computer Science — *CGPA: 7.44*

Sri Chaitanya Junior College, Tirupati — *2017*
Class XII (Intermediate) — *Percentage: 94.4%*

PROJECTS

Knee MRI Abnormality Detection — RSNA Kaggle Competition *2026*

PyTorch, timm (ConvNeXt), Hugging Face Transformers (Qwen2.5-14B, 4-bit), pydicom, scikit-learn

- Multi-label detection of **12 knee abnormalities** from multi-series MRI (≈820K DICOM files, 570 GB); only 58 of 4,407 studies had radiologist labels.
- Graded multilingual radiology reports with an **LLM**, calibrated and blended label sources, lifting label quality from 0.71 to **0.93 macro-AUC**.
- Built a 2.5D ConvNeXt with cross-series transformer attention; a 15-model TTA ensemble reached **0.906 private-leaderboard AUC** among 6,425 teams.

KIC-COCO: Kannada Image Captioning Dataset and Benchmark *2026*

PyTorch, IndicTrans2-1B, CLIP, ViT, Swin, ConvNeXt, IndicBART, BERTScore, sacreBLEU

- Constructed a **250K-caption Kannada corpus** over 50K MS COCO images with fixed splits; benchmarked **6 encoder–decoder architectures** (CNN–RNN with attention and transformer decoders).
- Best model ConvNeXtTransformer reached character-level **BLEU-4 68.65**, chrF 43.20 and back-translation **BERTScore F1 84.02**; paper under review at ICMLDE 2026.

AgroVision-7: Fine-Grained Rice Leaf Disease Recognition *2026*

PyTorch, ViT, Swin, ConvNeXt, Optuna, Focal Loss, TTA, Grad-CAM

- 7-class classifier on 14,241 images; ensemble of ViT-B/16, Swin-T and ConvNeXt-Tiny achieved **97.48% accuracy and 0.9750 macro-F1**, every per-class accuracy ≥ 0.96.
- Imbalance-aware pipeline (weighted sampler + class-weighted focal loss) lifted ResNet-50 by **+17.23 points**; 15-trial Optuna search, 5-view TTA and Grad-CAM explainability.

COD Return-Risk Scorer — Razorpay AI Buildathon 2026 *2026*

CatBoost, scikit-learn, SHAP, FastAPI, Docker, pytest

- Cost-sensitive, leakage-controlled risk scorer for Indian cash-on-delivery orders, deployed as a **live FastAPI service** with SHAP reasons per prediction.
- Compared 11 models with paired bootstrap CIs and set a cost-optimal threshold; saves **₹29,481 per 10K orders** (ROC-AUC 0.806).

SustainSched: Carbon-Aware Kubernetes Scheduling Agent *2026*

LangGraph, Ollama (Llama 3.2 3B), XGBoost, LSTM/GRU, FastAPI, Kubernetes, Prometheus, Grafana

- Autonomous agent that ranks 14 UK grid regions by forecast carbon intensity and emits explainable deploy / delay / migrate decisions.
- XGBoost forecaster beat LSTM and GRU with **R^2 0.917** (MAE 7.16); runs locally on a 6 GB laptop GPU.

Sentient: Self-Verifying AI Intelligence Agent *2026*

Next.js, TypeScript, Groq (qwen3-32b), Gemini embeddings, LanceDB, D3.js

- Agentic RAG over 8 live sources with a draft → verify → revise loop: a critic checks every claim against retrieved chunks before publishing a cited daily briefing.

EV Battery Analytics and AppealIQ Health-Plan Analytics *2026*

DuckDB, XGBoost, LightGBM, Optuna, SHAP, Isolation Forest, Power BI, Ollama

- EV: stacked virtual state-of-charge sensor with **2.1% error (R^2 0.96)** on 70 real BMW i3 drives, plus a local RAG + text-to-SQL assistant.
- AppealIQ: overturn-risk model on 42,710 real medical-review decisions, **ROC-AUC 0.736** on unseen years; NLP routing at 89.1% accuracy.

Mental Health Risk and Treatment Prediction with Agentic AI Integration *2025*

scikit-learn, XGBoost, LightGBM, SHAP, Optuna, Stacking Ensemble

- Benchmarked 11 classifiers; a stacking ensemble reached **95.22% accuracy and 0.9939 ROC-AUC**, with SHAP feature-importance consensus of 96% across models.

HACKATHONS & COMPETITIONS

Amazon ML Challenge 2026 — finished in the **top 3%** of the leaderboard. *2026*

Razorpay AI Buildathon 2026 — Track 02, AI Risk Manager: built and deployed the COD Return-Risk Scorer. *2026*

RSNA Knee Abnormality Detection (Kaggle) — 0.906 private AUC in a 6,425-team research code competition. *2026*

PUBLICATIONS

Wildfire Occurrence Analysis Using Meteorological Data: A Comparative Study of Statistical Methods and Learning Models *Apr 2026*

9th International Conference on Inventive Computation Technologies (ICICT 2026) — Published

KIC-COCO: A Benchmark Dataset for Kannada Image Captioning and Architecture Comparison *2026*

5th International Conference on Machine Learning and Data Engineering (ICMLDE 2026) — Under review

CERTIFICATIONS

AWS Certified Cloud Practitioner — Amazon Web Services
AWS Academy Graduate – Cloud Foundations — Amazon Web Services *Feb 2026*
Deep Learning Onramp; Image Processing Onramp — MathWorks